

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		award (1) for any of following no magnesium remains all the magnesium reacts no silver/grey solid remains only copper can be seen only brown solid can be seen solution is still blue accept all the magnesium dissolves / disappears		1		1		1
		(ii)		award (2) for all points plotted correctly – tolerance $\pm\frac{1}{2}$ small square award (1) for any 3 points plotted correctly award (1) for straight line		3		3	3	
		(iii)	I	0.07 ecf possible from incorrectly plotted graph			1	1	1	
			II	award (1) for any sensible method e.g. find mass of copper formed when 0.10g of magnesium is added and multiply value by 3 find mass of copper formed when 0.15g of magnesium is added and multiply value by 2 find the mass of copper formed when 0.10g and 0.20g of magnesium are added and add values			1	1	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		0.0200 (2) if answer incorrect award (1) for $M_r(\text{CuSO}_4) = 159.5$		2		2	2	2
		(ii)		0.0400 (2) ecf possible from part (i) if answer incorrect award (1) for volume in dm^3 0.5 $\frac{500}{1000}$		2		2	2	2
				Question 7 total	0	8	2	10	9	5